

YIFANG (MARK) CHEN

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EDUCATION

New York University Shanghai Shanghai, China
B.S. in Computer Science; Minor in Mathematics 2023.09 - 2027.05

Cumulative GPA: 4.0/4.0 | GRE: 331 (Q170, V161) | TOEFL iBT: 5.5/6.0 (110/120; Speaking 5.5/6.0, Writing 6.0/6.0)

- Awards: BaoSteel Outstanding Student Scholarship (**Top 0.25% across all majors and class years**); NYU Shanghai Recognition Award (**Top 4%**); Dean's List for Academic Year; Outstanding Student Worker of the Year
- Major Courses: Machine Learning (A), Databases (A), Probability & Statistics (A), Multivariable Calculus (A)

New York University New York, USA
Study Away Student in Computer Science 2025.09 - 2026.05

- Major Courses: Natural Language Processing (A; Graduate Level), Numerical Analysis (A), Parallel Computing (A)

PUBLICATIONS

[1] Jiayi Yang*, **Yifang Chen***, Yuanfu Sun, Jiajin Liu, and Qiaoyu Tan. "One Model, Many Graphs: Learning over Attributed Graphs across Heterogeneous Modalities with Vision-Language Models." Accepted to **EMNLP 2026 Main Conference**. arXiv:2607.19128. (*Equal contribution) [Paper](#)

RESEARCH EXPERIENCE

Generalizable Search Agents for General and Deep Search Shanghai, China
Project Lead; Co-advised by Prof. [Qiaoyu Tan](#) and Prof. [Jinyang Li](#) 2026.05 - Present

- Identified a generalization gap between agents optimized for retrieval-based question answering and long-horizon deep search, with deep-search agents underperforming on conventional search benchmarks.
- Explored training strategies combining **general-search and deep-search tasks** through supervised fine-tuning and RL-based policy optimization to adapt query reformulation, search depth, and stopping behavior to task difficulty.
- Built a controlled evaluation framework across PopQA, 2WikiMultiHopQA, BrowseComp, and BrowseComp-ZH, tracking **cross-regime accuracy, retrieval success, tool efficiency, and duplicate queries** under matched settings.

Multi-Agent Graph Learning with Adaptive Coordination Shanghai, China
Manuscript in preparation for ICLR 2027; Advised by Prof. [Qiaoyu Tan](#) 2026.05 - Present

- Formulated **multi-agent graph learning** as coordinated reasoning across four specialists covering local structure, extended neighborhoods, global graph context, and node semantics.
- Implemented and systematically evaluated **centralized, decentralized, and hierarchical collaboration strategies** for node classification and link prediction, revealing when specialist complementarity yielded gains.
- Proposed training-free and trainable **adaptive routers** that outperformed communication-heavy baselines while selecting instance-relevant specialists with 1-2 model calls versus 4-8 calls for communication-heavy baselines.

OMG-VLM: One Model, Many Graphs with Vision-Language Models Shanghai, China
Co-First Author; Advised by Prof. [Qiaoyu Tan](#) 2026.01 - 2026.05

- Developed a unified generative framework for reasoning across **text-, image-, and multimodal-attributed graphs** using a pretrained vision-language model as a shared backbone; designed target-aware textual retrieval and center-conditioned visual neighborhood adapters in the model's native token space.
- Outperformed all evaluated baselines across **9 in-domain and cross-graph transfer benchmarks**, achieving **88.69% average in-domain accuracy** and **66.67% average transfer accuracy**, with improvements of 3.38 and 15.09 percentage points, respectively, and gains of up to 20.15 points on unseen graphs.
- Built end-to-end training, evaluation, and ablation pipelines on HPC clusters covering **9 datasets and 10 representative baselines**; evaluated graph adapters, token budgets, neighborhood scaling, and optimization strategies for the co-first-authored paper accepted to **EMNLP 2026 Main Conference**.

EXTRACURRICULAR EXPERIENCE

myGenius, Exam Preparation Platform Powered by Generative AI Remote
Backend Developer 2024.09 - 2024.12

- Developed a back-end system using Python and PDF-lib to preprocess user-uploaded PDFs into structured formats for AI-driven question generation, facilitating the creation of mock exam content from course materials.
- Engineered dynamic and customizable AI prompts for OpenAI's GPT-4 API, optimizing question relevance and alignment with user-selected difficulty levels to ensure the generation of relevant multiple-choice questions.
- Developed an optimized MongoDB system for efficient storage and retrieval of AI-generated content, facilitating seamless integration with a web interface for displaying multiple-choice questions under user-specific queries.

TAMID Group, NYU Shanghai Shanghai, China
Principal of Tech Consultants 2024.02 - 2024.12

- Designed and coordinated a 15-session curriculum in web development and machine learning for 50+ participants.

SKILLS

- Python (PyTorch, Hugging Face, scikit-learn, NumPy), C, CUDA, MPI/OpenMP, Linux, Git, MySQL, LaTeX